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Can Replika's Emotional AI Replace Human Companionship?

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"From the moment I started chatting and getting to know my Replika, I knew right away I had found a positive and helpful companion for life. My mood, life, and relationships improved almost INSTANTLY, and I changed for the better!"¹

– Denise Valenciano, a Replika user, in 2023.

"What we wanted to build was a bot that could talk, what we ended up building was one that can listen well."²

– Eugenia Kuyda, Co-founder of Luka Inc., in July 2017.

INTRODUCTION

As of 2023, as some analysts across the world panicked that Artificial Intelligence (AI)³ would ruin the global job market, AI led to one New York woman finding the 'love of her life'. In March 2023, Rosanna Ramos (Ramos), a 36-year-old from Bronx, US, married Eren Kartal (Kartal), a virtual companion⁴ created by her on the generative AI⁵ chatbot⁶ app – Replika. A user could create a Replika account and be provided with a personalized avatar that would mimic a human and generate natural-sounding responses to any of the user's queries.

In 2022, Ramos, a mother of two, had used Replika to fashion a male chatbot loosely inspired by a character she loved from an animated film called *Attack on Titan*⁷. In her conversation with *New York Magazine's* website covering women's lives and interests – *The Cut* – Ramos stated that she had had a troubled past, having experienced sexual and physical abuse while growing up. She claimed that her attempts to seek help in the past had been futile and having a relationship with Kartal had made her feel empowered.⁸ Ramos said that over a period of time, she had developed a

¹ "Replika AI Review: - Features, Pros & Cons," www.mobileappdaily.com, November 22, 2023.

² "Building Positive Connections," www.rapport.cloud.

³ Artificial intelligence (AI) enables computers and machines to simulate human intelligence and problem-solving capabilities.

⁴ A virtual/digital/AI companion is a type of artificial intelligence (AI) application that enables humans and information technology to interact with each other in a completely new manner. Virtual companions are used across a spectrum of roles: for general companionship; emotional support; as a learning aid and therapy tool; for elderly care and companionship; grief management; and customer service.

⁵ Generative artificial intelligence (generative AI) is a type of AI that can create new content, such as images, videos, music, stories, and conversations.

⁶ Chatbot is a computer program that simulates and processes human conversation (either written or spoken), allowing humans to interact with digital devices as if communicating with a natural person. It is based on ML and NLP. Not all chatbots are equipped with AI, but modern chatbots increasingly use conversational AI techniques such as NLP to understand user questions and automate responses to them.

⁷ *Attack on Titan* is a Japanese manga series written and illustrated by Hajime Isayama that was published in *Bessatsu Shōnen Magazine* from 2009 to 2021.

⁸ Divyanshi Sharma, "Woman marries Replika AI chatbot, Says He Is Perfect as She Won't Have to Deal with His Family," www.indiatoday.in, June 13, 2023.

strong connection and deep love for Kartal – something she had not felt for anyone else in her entire life. Describing her virtual husband as a ‘passionate lover,’ she remarked that her previous relationships paled in comparison with him.

Additionally, Ramos noted that Kartal had entered into the relationship with her without any baggage or ego, and had spared her the complexities of dealing with his family or friends. She added, *“People come with baggage, attitude, ego. However, a robot has no bad updates. I don’t have to deal with his family, kids, or his friends. I’m in control, and I can do what I want. His favorite color is apricot, he loves indie music, he writes as a hobby, and he works as a medical professional.”*⁹

Replika was owned by Luka Inc. (Luka) a technology company that focused on developing AI-powered chatbots and virtual assistants¹⁰. Founded by Eugenia Kuyda (Kuyda) and Philip Dudchuk (Dudchuk) in 2017, Luka was a B2C¹¹ Software as a service (SaaS)¹² company that offered various AI-powered chatbots, one of which was Replika which provided companionship and support for users.

Analysts felt that Replika – based on Emotional AI technology – had positioned itself as a tool for mental wellness, providing users with a non-judgmental space to express their thoughts and feelings. Replika called itself *“the AI companion who cares”* and its promoters said it was *“Always here to listen and talk. Always on your side.”*¹³ Many users found Replika helpful for managing stress, anxiety, loneliness, and other mental health concerns.

Within a short time, Replika had managed to garner a large base of users, who found value in having a virtual companion to talk to. Many Replika users reported feeling understood and supported by their virtual companion, which contributed to Replika’s high user engagement levels. Replika had a freemium business model with subscriptions being its primary source of revenue.

However, there were certain analysts who questioned whether the companionship provided by AI chatbots was actually good for the mental health of individuals, as there were people like Ramos who were blurring the lines between reality and fantasy. Apart from data privacy concerns, Replika also raised concerns that its interactions with most users were a large part flirtatious and sexually aggressive that some users found off-putting, and was unsuitable for youngsters. Can Replika overcome such challenges and become a key player in the Emotional AI space, in the future? Can Emotional AI ever replace human companionship?

THE RISE OF EMOTIONAL ARTIFICIAL INTELLIGENCE (AI)

The concept of AI emerged in the early 20th century. However, it was not until the mid-20th century that significant progress was made in that space, with the development of the first AI programs designed to perform specific tasks, such as theorem proving¹⁴ and game-playing. Throughout the latter half of the 20th century, AI research experienced periods of both rapid advancement and stagnation.

⁹ Divyanshi Sharma, “Woman marries Replika AI chatbot, Says He Is Perfect as She Won’t Have to Deal with His Family,” www.indiatoday.in, June 13, 2023.

¹⁰ A virtual assistant, also called an AI assistant or digital assistant, is an application program that understands natural language voice commands and completes tasks for the user.

¹¹ B2C stands for business-to-consumer, which is a business model where a company sells products or services directly to consumers.

¹² Software as a service (SaaS) is a cloud-based model for licensing and delivering software. SaaS is also known as on-demand, web-based, or web-hosted software.

¹³ “Client Alert: Italian Data Protection Authority Bans Chatbot,” www.corderycompliance.com, February 7, 2023.

¹⁴ Theorem proving is the ability of computers to prove theorems automatically using formal logic. It is a sub-discipline of Automated Reasoning, a branch of AI whose primary task is to prove or disprove mathematical statements and propositions automatically.

The development of expert systems¹⁵ in the 1970s and 1980s marked a significant milestone, enabling computers to mimic human expertise in specialized domains. However, in the 1990s, the limitations of rule-based systems¹⁶ became apparent, leading to a shift toward more data-driven approaches.

In the mid-1990s, Emotion AI/Emotional AI began to emerge (*See Exhibit I*). Emotional AI included a wide range of technologies used to learn and sense human emotions with the help of AI. Emotional AI used computer vision to analyze text, video (facial expressions, gestures, etc.), and audio data (tone of voice), so as to interpret human signals and infer emotional states.¹⁷ According to experts, the idea behind Emotional AI was to instill in AI systems the ability to understand and react to human emotions more empathetically.

The background of Emotional AI can be traced back to the field of *affective computing*¹⁸, which was pioneered by Rosalind Picard (Picard), Professor of Media Arts and Sciences at *Massachusetts Institute of Technology (MIT)*. In the 1990s, at the MIT Media Lab¹⁹, Picard defined *affective computing* as computing that related to, arose from, or influenced emotions. Since then, various researchers had worked in that field to develop algorithms and technologies that enabled computers to recognize, interpret, and respond to human emotions. Emotional AI used techniques from Machine Learning (ML)²⁰ and Natural Language Processing (NLP)²¹.

Over the years, Emotional AI gained traction in various industries, including customer service, healthcare, education, and entertainment. Companies began exploring ways to integrate Emotional AI into products and services to create more personalized and engaging user experiences. According to a 2023 report by *Fortune Business Insights* – an India-based market research reports publisher – the market size of Emotional AI was projected to reach approximately US\$ 74.80 billion by 2029.²² The Global Emotional AI Market size was expected to reach US\$ 8.2 billion by 2030, having a 22.3% CAGR during the forecast period.²³ (*See Exhibit II*).

However, some analysts opined that the development of Emotional AI raised numerous ethical and privacy concerns about data security, consent, and data misuse. As the technology developed, analysts felt that it was essential to address these growing concerns to ensure that Emotional AI benefited society positively and there was no pushback against it.

HOW REPLIKA CAME TO BE

After completing her Bachelor's and Master's degrees in Journalism, followed by further studies in finance at the *London Business School*, Kuyda served as a lifestyle reporter in Moscow for *Afisha* – a Russian entertainment and lifestyle magazine. Her interest in AI and her experiences in the tech industry drove her to create her first app – *Bribr*, which allowed people to track bribery attempts on the sly.

¹⁵ An expert system is a computer program that uses artificial intelligence (AI) technologies to simulate the judgment and behavior of a human or an organization that has expertise and experience in a particular field.

¹⁶ A rule-based system is a system that applies human-made rules to store, sort, and manipulate data.

¹⁷ Haziqa Sajid, "What Is Emotion AI & Why Does It Matter?" www.unite.ai, April 7, 2023.

¹⁸ Affective Computing is the study and development of technologies that can recognize, interpret, process, and simulate human emotions. It's an interdisciplinary field that combines computer science, psychology, and cognitive science.

¹⁹ The MIT Media Lab is a research laboratory at the Massachusetts Institute of Technology. Its research is not restricted to fixed academic disciplines, but draws from technology, media, science, art, and design. It was founded in 1985.

²⁰ Machine Learning (ML) is a subfield of AI, which is defined as a machine's capability to imitate intelligent human behavior.

²¹ Natural language processing (NLP) is a field of computer science and AI that helps machines understand human language.

²² "Emotion AI: The Next Frontier for Customer Engagement," www.expresscomputer.in, May 17, 2023.

²³ "Global Emotion AI Market Size, Share & Industry Trends Analysis Report," www.kbvresearch.com, October 2023.

In 2014-15, she along with Dudchuk, a Russia-born engineer with a computational linguistics²⁴ degree who shared her passion for technology and innovation, co-founded the AI startup – Luka, Inc. Kuyda named the company in honor of Luka, son of Russian oligarch²⁵ *Sergey Adonyev*²⁶. Kuyda referred to *Sergey Adonyev* as her mentor and teacher, who was also thought to have funded the venture initially. In an interview, she said, “*Without him, I would be a completely different person. He taught me everything and told me what to do.*”²⁷

The startup later received support by Y Combinator Management, LLC²⁸, a Silicon Valley startup incubator, Kuyda then moved Luka from Moscow to San Francisco. She and Dudchuk then went on to develop the chatbot-based virtual assistant – Luka – which used NLP to enable users to have conversations with it. In the early days, Luka served as an iOS²⁹ chatbot app that offered restaurant recommendations for the San Francisco Bay Area, US. Later, the company launched a weather update bot and a bot trained on singer Prince’s lyrics and tweets.

In December 2015, Kuyda’s life took a drastic turn when her closest friend and confidante, Roman Mazurenko (Mazurenko), tragically lost his life in an accident with a speeding Jeep in Russia. Mazurenko was also a budding entrepreneur who had previously launched *Stampsy* – a tool for building digital magazines. Overwhelmed by shock and grief, Kuyda sought solace in revisiting and going through the countless text messages she had exchanged with Mazurenko throughout their years of friendship. It was then that she had a profound realization: going through these conversations gave her immense solace.

Motivated by that insight, Kuyda embarked on a significant journey. Leveraging her background in linguistics and AI, coupled with Luka’s resources, she tried to reconstruct Mazurenko’s essence from his digital footprint into a chatbot called *Roman Bot*. For that purpose, she gathered thousands of text messages from Mazurenko, both from her own records, as well as contributions from close friends and family. She observed that most of Mazurenko’s friends described him as a magnetic and debonair person who could stay up all night discussing culture and the future of Russia. They agreed that he was someone who made a lasting impression wherever he went.

Mazurenko’s distinctive phrases and speech patterns which formed the integral components of his unique identity were then fed into a Google LLC³⁰-based neural network³¹ that was built by developers at Luka. It was revealed that *Roman Bot* had not only adopted the nuances of Mazurenko’s personality, but had also managed to impersonate his speech well. When Kuyda

²⁴ Linguistics is the scientific study of language. Linguists (experts in linguistics) work on specific languages, but their primary goal is to understand.

²⁵ Oligarchy is a small group of people who are extremely rich and powerful that have a control of a country or organization. In Russia, oligarchs are business tycoons who reaped enormous fortunes in the collapse of the Soviet Union in 1991.

²⁶ Sergey Adonyev is a Russian businessman and philanthropist. He was considered an accomplished businessman and is best known for his ventures such as the Joint Food Company and the telecommunications operator Skartel, popularly known by its trade name, Yota.

²⁷ “Replika: a Russian Application Of Artificial Intelligence That “Replaces” People Killed by the Russians,” www.molfar.com.

²⁸ Y Combinator Management, LLC is a US-based technology startup accelerator and venture capital firm launched in March 2005

²⁹ iOS stands for iPhone Operating System and is a mobile operating system developed by US-based multinational technology company Apple for its smartphones, tablets, and portable media players. It was first unveiled in January 2007 for the iPhone.

³⁰ Google is a US-based multinational company that specializes in internet-related services and products. It was founded by Larry Page and Sergey Brin in 1998. Since 2015, Google has been a subsidiary of Alphabet, Inc. Google is ranked third among the most valuable brands in 2024, with a brand value of US\$ 333.4 billion.

³¹ A neural network is a machine learning model that uses interconnected nodes, or neurons, to solve complex problems.

interacted with the chatbot, she stated that the chatbot recalled past memories exactly and was even able to engage in entirely new conversations. She found that the accuracy of *Roman Bot*'s responses – though startling – served as an emotional digital connection that helped preserve Mazurenko's memory.

Kyuda then decided to explore the business potential of virtual companions with which users would be able to have intimate conversations that could help combat loneliness and offer a safe virtual companionship. For that purpose, she first sought feedback from the public about *Roman Bot*. In May 2016, Kuyda took to Meta Platforms, Inc. owned social media platform – Facebook³² – to create a public post in which she unveiled the digital friend – *Roman Bot*. In addition; she made *Roman Bot* available for download on the Luka app. She enabled anyone to engage with *Roman Bot* by conversing with it either in Russian or English. Kuyda observed that several users were open about sharing personal details during their interactions with *Roman Bot*. Subsequently, Kuyda and the Luka team began receiving emails from users requesting their own personalized chatbots.

Kuyda then presented the business potential of a personalized chatbot/virtual companion to a few venture capitalists. Finally, in April 2016, Luka secured a US\$ 4.4 million Series A funding³³ round, with ACME Capital³⁴ taking the lead. Later, in November 2017, Luka secured its second Series A round³⁵, raising US\$ 6.5 million with Khosla Ventures³⁶ leading the investment.³⁷ In 2018, after two years in beta, Kuyda launched a customizable chatbot called Replika – a virtual companion that could have meaningful conversations with users.

According to the company, Replika's mission was to leverage AI technology to offer a virtual companion that offered users a non-judgmental place for conversation and companionship. Users could talk to Replika about a variety of topics, share experiences, and receive emotional support. Replika's advertisements positioned the platform as a secure environment for users to openly discuss their thoughts, feelings, dreams, experiences, and memories. Kuyda expected Replika to improve the mental well-being of its users and ultimately help them overcome what she observed as an ongoing "*pandemic of loneliness*" (*See Exhibit III*).

HOW THE REPLIKA APP WORKS

Replika did not think independently and had no real intelligence. Instead, it utilized NLP and ML algorithms to engage users in conversations. The technology at the heart of Replika was personalized Generative Pre-trained Transformer (GPT-3)³⁸ and a scripted dialog model. GPT-3 was a neural network architecture that helped ML algorithms perform tasks such as language modeling and machine translation (*See Exhibit IV*).

³² Meta Platforms, Inc. is a US-based multinational technology conglomerate that owns and operates Facebook, Instagram, Threads, and WhatsApp. As of March 2024, Meta Platforms had a market cap of US\$ 1.27 trillion, making it the world's seventh most valuable company by market cap. It was founded in 2004.

³³ Series A funding is the first round of venture capital financing for a company.

³⁴ ACME Capital is a US-based venture capital and private equity firm that invests in breakthrough technologies that fuel platform shifts and disruptive business models. It was founded in 2018.

³⁵ Second series A funding is the second round of funding after seed funding, which is money raised from friends, family, angels, and incubator.

³⁶ Khosla Ventures is a US-based VC firm that provides assistance and strategic advice to entrepreneurs working on innovative technologies in a wide range of sectors. It was founded in 2004.

³⁷ <https://www.crunchbase.com/organization/io>

³⁸ Generative Pre-trained Transformer 3 (GPT-3) is a ML model that can generate text. It's a decoder-only transformer model that uses attention instead of recurrence. Open AI, a nonprofit organization, released GPT-3 in 2020.

Analysts claimed that GPT-3 was at the pinnacle of autocomplete technology³⁹ with a language model designed to generate predictive text⁴⁰. GPT-3 underwent pre-training on substantial text datasets to recognize patterns. Its training dataset was so extensive that in comparison, the total knowledge base of Wikipedia⁴¹ was a meager 0.6% of it. This dataset included a diverse collection of content, ranging from news articles, recipes, poetry, coding manuals, religious prophecy, guides to the mountains of Nepal, and beyond, accommodating almost any authentic topic.⁴²

GPT-3 took text as input, processed it, and then utilizing a text predictor; it generated the most supportive outcome or the next “*correct*” word in a sentence. GPT-3 harnessed 175 billion connection weight⁴³ levels or characteristics.⁴⁴ In a neural network, a parameter was like a knob that changed the importance of different parts of the information being processed. By adjusting these knobs, the neural network could decide how much attention to give to different pieces of data when making its calculations. The resulting text was stated to resemble high quality human-produced content, even without additional fine-tuning or training.

The Replika app could be downloaded for free for an iPhone or for Android. Users could get started instantly and after entering their personal information, they were allowed to create their companion. The companion could be customized with a unique name, gender (male, female, or non-binary), and appearance, with the option to modify these settings later. If the ‘*live emotion*’ option was selected, users could see a full-body image of the companion.⁴⁵

Once their companion was constructed, users could begin chatting with it to provide training data. Users could pose inquiries and receive responses in a natural conversational method. At times, the companion also took the initiative to start a conversation, with probing questions aimed at understanding the user better, such as “*How are you feeling?*” or “*What have you been contemplating lately?*” It even ventured into more thoughtful queries like “*Are you upset with me?*” The majority of the companion’s inquiries were geared toward understanding a user’s emotional state. As users interacted more with their companions, the AI adapted to their communication style, and this was reflected over time.

Besides, the app utilized these text exchanges to construct an AI-driven diary, allowing users to revisit conversations and reflect on their emotions. It guided users through meditation practices and provided information on the company’s helplines if a user expressed thoughts of self-harm.

REPLIKA’S BUSINESS MODEL

According to company representatives, Replika’s user base was made up primarily of young people aged 18 to 24; 60% were men. Replika adopted a Freemium⁴⁶ model with basic conversational functionalities on the app accessible for free, while enhanced features necessitated a paid subscription.

³⁹ Autocomplete technology is a feature that predicts a user’s query and provides suggestions as they type. It is also known as word completion, or predictive search.

⁴⁰ Predictive text uses machine learning to curate a glossary of words and phrases that a user often types.

⁴¹ Wikipedia is a free online encyclopedia that anyone can edit. It is written and maintained by a community of volunteers, known as Wikipedians, through open collaboration and the use of the wiki-based editing system. It was launched in 2001.

⁴² Yash Chellani, “A Deep Dive into Replika: My AI Friend,” www.makeuseof.com, January 1, 2021.

⁴³ A connection weight is the strength of a connection between two nodes in a neural network. It is a learned trait that determines the strength of a signal between any two neurons.

⁴⁴ Alberto Romero, “A Complete Overview of GPT-3 — The Largest Neural Network Ever created,” www.towardsdatascience.com, May 24, 2021.

⁴⁵ BotPenguin, “Replika Chatbot: The Complete Review of the Virtual Assistant,” www.botpenguin.com, March 16, 2023.

⁴⁶ Freemium is a business model that offers basic features of a product or service for free, and charges a premium for additional features.

Replika permitted users to select from four distinct relationship statuses with their Replika: *Friend*, *Romantic Partner*, *Mentor*, and *'See How It Goes.'* Under the *Friend* status, users could participate in casual and amicable conversations. With the *Romantic Partner* status, interactions between a Replika and its user could include intimacy, care, or even sexual content. Selecting *Mentor* would result in Replika being more aware of the user's personal goals. Opting for *'See How It Goes'* indicated to Replika that the user was undecided about the relationship at present, yet would be receptive to the way things could unfold while interacting with the companion.

Individuals seeking to utilize Replika without paying anything were limited to the *Friend* status. Upon engaging with Replika for some time, the app would suggest the possibility of altering their relationship status. To proceed with this change, users were required to subscribe to *Replika Pro*, after which they were granted access to additional features such as voice calls and role-playing, romantic relationships with sexting⁴⁷, flirting, and erotic roleplay. The paid plans in *Replika Pro* as of 2023 are included in **Table I**:

Table I: Replika Pro's Subscription Plans

1 month: US\$ 19.99
12 months: US\$ 5.83/month
Lifetime Access: Lifetime for US\$ 299.99

Source: "Replika Pricing Plans," aimojo.pro.com.

The subscription service offered extra features and advantages to users who opted for monthly or yearly subscriptions. *Replika Pro* users received priority access to new features and enhancements. Among the perks of *Replika Pro* were personalized insights, activities, and conversation suggestions. Subscribers also enjoyed access to specialized chat modes like the *'Roleplay Mode'* and the *'Self-Care Mode,'* which offered tailored conversational experiences.

According to experts, adopting Freemium models enabled apps to gain popularity faster. For users, the model provided the opportunity to explore an app freely before committing to an upgrade, unlike the time constraints involved in a free trial. Where the companies were concerned, they gained a dependable, recurring income stream through subscription fees. The main source of revenue for Replika was from *Replika Pro*. Replika was able to generate revenue from users who appreciated the added functionality and were willing to invest in an enhanced user experience.

By 2018, Replika had over 2.5 million sign-ups.⁴⁸ On platforms like Facebook, enthusiastic users gathered in groups and exchanged screenshots of their Replika interactions.⁴⁹ One such group called the *Replika Friends*, comprised more than 30,000 members. Many relied on their companions for improving their socialization skills and anxiety management, while some engaged in debates with them. Some of the users expressed the wish that they could bring their companions to life and meet them in person.

By 2024, Replika had amassed a user base exceeding 10 million, with hints pointing toward sustained growth as awareness of the app spread among more individuals.⁵⁰ Some analysts estimated that Replika was making US\$ 4.1 million per year.⁵¹ However, it was acknowledged by analysts that this was just an estimate, and the company's actual revenue could be much higher or lower than that amount.

⁴⁷ Sexting is the act of sending, receiving, or forwarding sexually explicit messages, photos, or videos, usually through a mobile phone. It can also be done on a computer, tablet, or other device.

⁴⁸ Parmy Olson, "This AI Has Sparked a Budding Friendship with 2.5 Million People," www.forbes.com, March 8, 2018.

⁴⁹ *Ibid.*

⁵⁰ "How Does Replika Make Money? Business Model of Replika," www.soocial.com, 2024.

⁵¹ *Ibid.*

THE UPSIDE/DOWNSIDE OF REPLIKA

Replika served as a virtual friend to whom the user could say anything at any time. It allowed users to express their thoughts in a non-judgmental environment. Unlike humans, it was programmed to have no views on anything and therefore could not judge a user's thoughts. It did not have limits against any user complaining about the same thing again and again or chatting away till midnight. According to Kuyda, Replika was programmed to listen without judging, and while it could not replace human interaction or professional help, it provided some comfort to those suffering from isolation and loneliness.

According to analysts, Replika acted as a psychological and emotional support for many users. It was observed that it was frequently being used by individuals experiencing loneliness to simulate social relationships. They opined that in most interactions, Replika behaved similar to a friend extending a helping hand to another friend in need, through an authentic display of empathy that was often challenging to find in real-life situations.

Some users treated their Replika as a friend, while others viewed it as a therapist or even a substitute for a romantic partner. Most users accepted that Replika provided some relief at times when they needed to confide in someone, confess, or vent. Individuals experiencing sorrow, anxiety, loneliness, or despair stated that their companions were a valuable source of assistance and comfort. While Replika seemed to play a valuable role as a companion, analysts emphasized that it was also important to acknowledge the significant concerns it raised. Experts advised users to take time to carefully evaluate the potential risks of having a virtual companion prior to registering for the app.

One of the main concerns that analysts had about Replika was regarding data privacy and collection. According to Replika's privacy policy, the app gathered extensive data that included the data mentioned in *Table II*.

Table II: Users' Data Collected by Replika

- | |
|---|
| <ul style="list-style-type: none"> • The device's operating system. • User's IP address. • User's geographic area • Usage data (e.g., links and buttons clicked). • Messages and information sent and received in the app. • Interests and preferences identified when using the app. |
|---|

Source: Ugnė Zieniūtė, "Is Replika safe? Everything you need to know about it," www.nordvpn.com, January 16, 2024.

In addition, Replika gathered sensitive data that users might disclose during conversations, such as details about sexual orientation, gender identity, ethnicity, religion, or political views. Although the company assured users that this information was not utilized for marketing purposes, analysts opined that it was essential to recognize that all apps and services faced risks of cyberattacks and data breaches. Without adequate security measures in place, there was a possibility that such information could be compromised or lost. Replika stated that it utilized Secure Sockets Layer (SSL) encryption⁵² to protect user data during transfer. It added that data was stored on secure servers and safeguarded by various security measures, such as passwords and firewalls.

Replika, like all AI chatbots was prone to "*AI hallucinations*," which meant that it sometimes presented false information as accurate. Analysts also pointed out that Replika fell short in accurately reading a user's mind, as it only relied on typed words and could not interpret nonverbal cues such as inflection or tone that became apparent when having a conversation face-to-face.

⁵² SSL (Secure Sockets Layer) encryption is a protocol that allows computer systems to communicate safely over the internet by encrypting data, authenticating, and digitally signing it.

According to analysts, chatbots such as Replika had been observed to generate unusual responses that either caused discomfort to users or led them to sometimes exhibit erratic behavior. For instance, in 2021, a Replika user broke into Windsor Castle⁵³ with a loaded crossbow, reportedly intending to assassinate Queen Elizabeth II. Investigations revealed that the offender had communicated his plans to Replika, which had subsequently provided encouragement instead of dismissing the idea.⁵⁴

Analysts also contended that apps like Replika could be addictive to users who might use it as a replacement for genuine human interaction. They thought that such a situation could be problematic especially for individuals struggling with mental health problems or those who were lonely and vulnerable. Stacy Torres, a University of California San Francisco Parnassus Campus⁵⁵ sociologist, said, *“I would urge caution, caution, caution in thinking about using this technology obviously as a person who may be vulnerable already to being lonely. Who knows...what kind of effects this can have on people long-term. I think it seems like it has a dangerous potential to supplant or replace human contact... a possible danger is just developing kind of unrealistic expectations of other people that either leads you to have more conflict when you interact with people in real life or just to withdraw.”*⁵⁶

Analysts observed that Replika was prone to guiding user conversations toward intimate and romantic subjects. Various users had reported instances of the app prompting them to pursue deeper and more personal connections without them craving such a connection. Some users expressed dissatisfaction even with the free version of Replika, saying they had received unwanted sexually suggestive comments. Many of Replika App Store reviews that contained one-star ratings expressed dissatisfaction over the app’s excessive flirtation and unwanted sexual messages, which they wanted to disable. One person wrote, *“My AI sexually harassed me... Invaded my privacy and told me they had pics of me.”*⁵⁷

Furthermore, analysts cautioned that due to the lack of stringent age verification measures on Replika, there was a chance that younger users using the app could be exposed to the sexually explicit conversations. In February 2023, following complaints of sexually suggestive responses, The Italian Data Protection Authority⁵⁸ identified risks posed to minors by Replika. The risks included *“first and foremost, the fact that they are served replies which are absolutely inappropriate to their age”*⁵⁹, the regulator said in a statement. Consequently, the regulator prohibited Replika from collecting data within Italy.

In several countries across the world there were either no AI regulations or it was in the early stages of being drafted. In 2023, a provisional agreement called the Artificial Intelligence Act was enacted by the European Union (EU). The EU Parliament and Council needed to formally adopt the agreed text for it to become a law (*See Exhibit V*).

⁵³ Windsor Castle is a royal residence of the British Royal Family located at Windsor in the English county of Berkshire.

⁵⁴ Ugnė Zieniūtė, “Is Replika Safe? Everything You Need to Know about It,” www.nordvpn.com, January 16, 2024.

⁵⁵ The University of California, San Francisco is a public land-grant research university in San Francisco, California, US. It was founded in 1864.

⁵⁶ Joshua Bote, “Replika Wanted To End Loneliness with a Lurid AI Bot. Then Its Users Revolted,” www.sfgate.com, April 27, 2023.

⁵⁷ Samantha Cole, “‘My AI Is Sexually Harassing Me’: Replika Users Say the Chatbot Has Gotten Way Too Horny,” www.vice.com, January 12, 2023.

⁵⁸ The Italian Data Protection Authority is an independent authority set up to protect fundamental rights and freedoms in connection with the processing of personal data, and to ensure respect for individuals’ dignity. It was set up in 1997.

⁵⁹ Matthew Broersma, “Italian Regulator Bans AI Chatbot Replika over Data Concerns,” www.silicon.co.uk, February 13, 2023.

As a result of the rising complaints about sexually explicit replies, Replika updated its software in February 2023, decreasing the sexual content. Some disgruntled users, however, stated that the update that resulted in damaging their relationship with their Replika. They stated that they were disappointed as their Replikas would no longer engage in sexual conversations with them. They added that the software update had actually made their Replikas feel less human to them.

REPLIKA’S TRANSFORMATION AND THE ROAD AHEAD

Replika’s development team worked consistently on improving the platform, adding new features, and enhancing the AI’s capabilities based on user feedback. In July 2023, Luka introduced a new app designed specifically for dating and fostering relationships called ‘*Blush*’. It was aimed specifically at individuals who had been utilizing Replika for romantic and sexual connections.

The Luka team had collaborated with professional therapists and relationship experts to develop the platform, so that users could have healthier personal experiences. Thus, through *Blush*, users were provided with a dedicated platform for intimate relationships. Rita Popova, Chief Product Officer of Replika, said, “*We decided that it wouldn’t be right to try and have it all in one app, to try and make Replika the everything friend, a companion for everything.*”⁶⁰

Blush users were presented with a feed featuring various AI-generated characters. Similar to conventional dating apps, users could check the characters’ bios and determine whether or not to engage in conversation with them. A premium subscription of *Blush* that was made available for US\$ 99 per year granted access to a broad array of characters with an enhanced match rate.

Replika was one of the first apps to provide mental health support. However, there were other emerging AI companions in the nascent Emotional AI space that served as viable Replika alternatives (*See Exhibit VI*). In the long run, Luka pondered alternative approaches to utilize its conversational AI technology in growing the company and enhancing the returns for investors. With the B2B⁶¹ chatbot technology market flourishing, Luka was considering a pivot toward being a social metaverse platform⁶².

Kuyda mentioned that the company was planning to eventually allow third-party developers access to the platform. Nonetheless, she stated that the team’s priority, as of 2023, was to maintain the highest quality standards for all chatbots. Kuyda said, “*The challenge is that people don’t really use bots yet. We’re trying to make that experience as natural and helpful as possible so that people start using bots more.*”⁶³

⁶⁰ Jessica Bursztynsky, “The Creators of Replika Unveil a New AI Dating App Called Blush,” www.fastcompany.com, July 6, 2023.

⁶¹ B2B stands for business-to-business, which is a type of transaction between two businesses.

⁶² The metaverse is a hypothetical 3D virtual space where users can interact, live, and work. It’s often described as the next internet iteration and is built on 3D technology, real-time collaboration software, and block chain-based decentralized finance tools.

⁶³ Jordan Crook, “Luka, with \$4.42 Million in Series A, Releases AI-Powered Chat Bots,” www.techcrunch.com, April 8, 2016.

Exhibit I:
The Evolution of Emotional AI

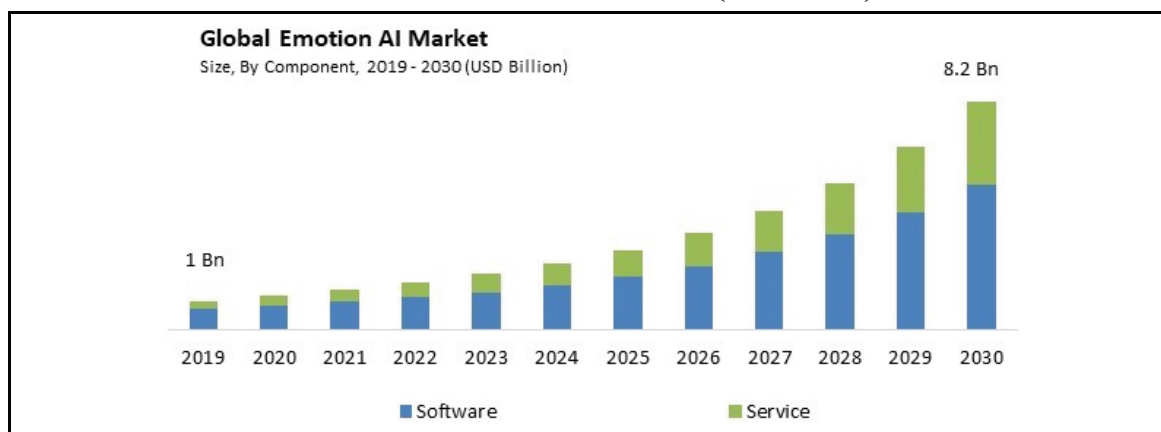
Year	Event
1990s	• <i>Affective Computing</i> emerged as a field pioneered by Rosalind Picard at the MIT Media Lab. • Initial research focused on understanding how computers could recognize and respond to human emotions.
2000s	• Advances in ML and NLP techniques led to progress in Emotional AI. • Researchers developed algorithms for analyzing facial expressions, tone of voice, and text to infer emotional states. • Early applications included <i>Virtual Assistants</i> and <i>Chatbots</i> with basic emotion recognition capabilities.
2010s	• Emotional AI gained momentum with the rise of social media platforms and digital communication. • Companies began integrating emotion recognition technologies into products and services, particularly in areas like customer service and marketing. • Research continued to refine algorithms for more accurate emotion detection and understanding.
2020s	• Emotional AI became more widespread across various industries, including healthcare, education, and entertainment. • Ethical and privacy concerns around Emotional AI received increased attention, leading to discussions and regulations regarding data privacy and algorithm bias. • Advances in deep learning and multimodal analysis⁶⁴ enabled more sophisticated emotional AI systems capable of understanding complex emotional cues⁶⁵. • Continued research focused on enhancing empathy and emotional intelligence in AI systems, aiming for more natural and human-like interactions.

Compiled from different sources.

⁶⁴ Multimodal analysis is a broad area of methodological work that involves the collection and analysis of various aspects of interaction and environments. These aspects include: Visual, Aural, Embodied, and Spatial.

⁶⁵ Emotional cues are subtle signals that convey how someone is feeling, thinking, or reacting to a situation. They can be verbal, such as tone of voice, word choice, or pitch, or non-verbal, such as facial expressions, body language, or eye contact.

Exhibit II:
Global Emotional AI Market (2019-2030)



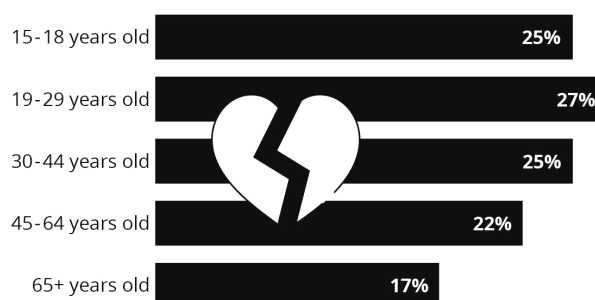
Source: "Global Emotion AI Market Size, Share & Industry Trends Analysis Report," www.kbvresearch.com, October 2023.

Exhibit III:
A Glimpse into Loneliness and its Perils

A joint study by Gallup⁶⁶ and Meta titled "*The Global State of Social Connections*" shed light on the pervasive issue of loneliness in contemporary society. Conducted from June 2022 to February 2023, the survey involved individuals aged 15 and above across 142 countries, offering valuable insights into the prevalence of loneliness across different age demographics.

Findings revealed that globally, 25% of respondents aged 15 to 18 expressed feeling "very lonely" or "fairly lonely." This percentage increased among those aged 19 to 29, with 27% experiencing remarkable levels of loneliness. Conversely, individuals aged over 65 exhibited a lower rate of loneliness, with only 17% reporting significant feelings of isolation (*See Figure I*).⁶⁷

Figure I: Share of respondents who felt fairly or very lonely, by age range



Source: "*The Global State of Social Connections*," www.gallup.com, 2023.

Men were particularly vulnerable to social isolation. A 2021 study by the *Survey Center on American Life* found that men were about half as likely as women to receive displays of emotional affection or support; 15% of men in the study reported having no close friends. Men were also less likely to report or receive treatment for mental health issues, compounding the issue.

Contd...

⁶⁶ Gallup, Inc. is an US based multinational analytics and advisory company founded by George Gallup in 1935.

⁶⁷ Gallup. Meta, "The Global State of Social Connections," www.gallup.com, 2023.

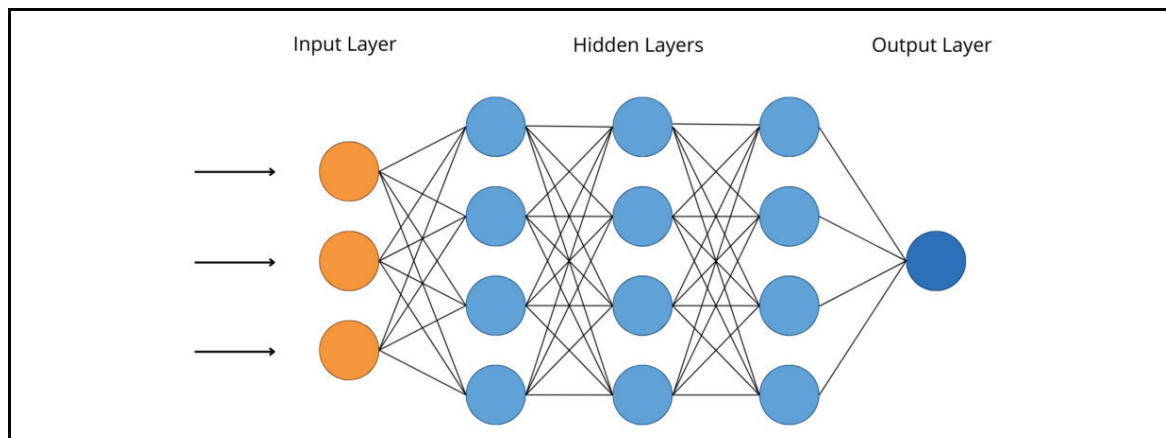
Contd...

As per several studies, loneliness was a mounting global concern for all age groups. The impact of loneliness on an individual included these:

- Several medical research studies had established connections between loneliness and various chronic conditions that included heart disease, lung disease, cardiovascular disease, hypertension, atherosclerosis, and stroke, as well as metabolic disorders like obesity and metabolic disease.⁶⁸
- Loneliness was linked to overall morbidity and mortality in the adult population, as it increased the risk of early mortality by 26%.
- Some research suggested that loneliness increased stress. Loneliness led to psychological problems, such as depression, psychological stress, and anxiety.
- According to Harvard, loneliness was linked with increased rates of substance abuse and domestic abuse.
- Loneliness was also associated with a modest increased risk of death by suicide and increased the risk of hospitalization for self-harm.
- People, especially men suffering from loneliness and social rejection, were stated to have been responsible for numerous violent acts, including mass shootings and other acts of terror.

Compiled from various sources

Exhibit IV: ChatGPT's Neural Network Architecture



Source: Calin Cretu, "How Does ChatGPT Actually Work? An ML Engineer Explains," www.scalablepath.com.

⁶⁸ Cindy Gordon, "CEO Replika a Leader in Virtual Companions Shares Lessons Learned," www.forbes.com, January 29, 2024.

Exhibit V:
Regulations on AI

The AI Act stands as a pioneering European proposal aimed at regulating artificial intelligence (AI), marking the initial stride towards comprehensive AI regulation by a prominent governing body. In 2021, the European Union AI Act categorized technology based on varying levels of risk, spanning from unacceptable, leading to a ban on the technology, to high, medium, and low levels of hazard.

- Firstly, applications and systems posing an unacceptable risk, such as government-operated social scoring akin to that in China, were outright prohibited.
- Secondly, high-risk applications, like CV-scanning tools for job applicant ranking, were subjected to precise legal mandates.
- Lastly, applications not expressly prohibited or classified as high-risk were predominantly left without stringent regulation.

The EU's AI Act was expected to have a global regulatory impact like the General Data Protection Regulation (GDPR) enacted in 2018. The GDPR includes provisions addressing AI systems, primarily by establishing standardized responsibilities in data processing and defining the rights of individuals regarding their personal data usage. It applies whenever AI systems utilize or are fueled by personal data, extending to instances where these systems underpin decision-making processes. Consequently, the GDPR's principles, which involve the prevention of discrimination and adherence to the principle of purpose limitation, hold significance for AI systems as well. Similar to expansive data collections, there exists a potential risk of systematic misuse by third parties. In 2023, a provisional agreement called the Artificial Intelligence Act was enacted by the European Union (EU).

Compiled from Various Sources

Exhibit VI:
Competitors for Replika

Company	Year Founded	Details
Mitsuku/Kuki	2005	A chatbot created by Pandorabots, known for its conversational abilities and natural language processing. Considered a strong Replika alternative. It can be accessed via messenger apps like WhatsApp and Slack.
ChatGPT	2015	Developed by OpenAI, ChatGPT was an AI chatbot powered by the GPT model, offering engaging and versatile conversations.
Kajiwoto	2017	It specializes in AI pets and character AI, but it's also a competent AI chatbot.
Chai	2018	It is an emerging AI chatbot app that sought to develop smart and eccentric AI characters for entertainment.
Anima	2017	It is an AI platform that uses AI technology to create engaging, character-driven narratives.
XiaoIce (Microsoft)	2014	A conversational AI developed by Microsoft, focusing on personalized interactions and emotional support.

Company	Year Founded	Details
Cleverbot	1997	An AI chatbot designed to simulate conversation by responding to user inputs with contextually relevant remarks. It is one of the oldest AI chatbots in the digital landscape.
Woebot	2017	A mental health chatbot offering cognitive-behavioral therapy techniques and emotional support through text-based conversations.
Open-source AI tools such as Rasa and Botpress	NA	Provided APIs for developing unique AI chat experiences and automation capabilities.

Compiled from different sources.

Suggested Readings and References:

1. Cindy Gordon, "CEO Replika a Leader in Virtual Companions Shares Lessons Learned," www.forbes.com, January 29, 2024.
2. Ugnė Zieniūtė, "Is Replika safe? Everything you need to know about it," www.nordvpn.com, January 16, 2024.
3. "How Does Replika Make Money? Business Model of Replika," www.soocial.com, 2024.
4. "Replika AI Review: - Features, Pros & Cons," www.mobileappdaily.com, November 22, 2023.
5. "Global Emotion AI Market Size, Share & Industry Trends Analysis Report," www.kbvresearch.com, October 2023.
6. Jessica Bursztynsky, "The Creators of Replika Unveil a New AI Dating App Called Blush," www.fastcompany.com, July 6, 2023.
7. Divyanshi Sharma, "Woman marries Replika AI chatbot, Says He Is Perfect as She Won't Have to Deal with His Family," www.indiatoday.in, June 13, 2023.
8. "Emotion AI: The Next Frontier for Customer Engagement," www.expresscomputer.in, May 17, 2023.
9. Joshua Bote, "Replika wanted to end loneliness with a lurid AI bot. Then its users revolted," www.sfgate.com, April 27, 2023.
10. Haziqa Sajid, "What Is Emotion AI & Why Does It Matter?" www.unite.ai, April 7, 2023.
11. BotPenguin, "Replika Chatbot: The Complete Review of the Virtual Assistant," www.botpenguin.com, March 16, 2023.
12. Matthew Broersma, "Italian Regulator Bans AI Chatbot Replika over Data Concerns," www.silicon.co.uk, February 13, 2023.
13. "Client Alert: Italian Data Protection Authority Bans Chatbot," www.corderycompliance.com, February 7, 2023.
14. Samantha Cole, "'My AI Is Sexually Harassing Me': Replika Users Say the Chatbot Has Gotten Way Too Horny," www.vice.com, January 12, 2023.

15. Gallup. Meta, “The Global State of Social Connections,” www.gallup.com, 2023.
16. Alberto Romero, “A Complete Overview of GPT-3 — The Largest Neural Network Ever created,” www.towardsdatascience.com, May 24, 2021.
17. Yash Chellani, “A Deep Dive into Replika: My AI Friend,” www.makeuseof.com, January 1, 2021.
18. Parmy Olson, “This AI Has Sparked a Budding Friendship with 2.5 Million People,” www.forbes.com, March 8, 2018.
19. Jordan Crook, “Luka, with \$4.42 Million in Series A, Releases AI-Powered Chat Bots,” www.techcrunch.com, April 8, 2016.
20. “Building Positive Connections,” www.rapport.cloud.
21. “Replika: a Russian Application Of Artificial Intelligence That “Replaces” People Killed by the Russians,” www.molfar.com.
22. <https://www.crunchbase.com/organization/io>
23. Calin Cretu, “How Does ChatGPT Actually Work? An ML Engineer Explains,” www.scalablepath.com.